Renewable Energy 227 (2024) 120483

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Numerical assessment and optimization of photovoltaic-based hydrogen-oxygen Co-production energy system: A machine learning and multi-objective strategy

Ningbo Wang, Yanhua Guo, Lu Liu, Shuangquan Shao *

School of Energy and Power Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China

A R T I C L E I N F O

Keywords:
Hydrogen-oxygen co-production
Solar photovoltaic
PEM electrolysis
Machine learning
Multi-objective optimization

A B S T R A C T

Green hydrogen production technology based on photovoltaic (PV), battery energy storage system (BESS) and proton exchange membrane (PEM) water electrolysis plays a crucial part in the transition process of energy to zero carbon technology. But few studies have focused on achieving all-day continuous hydrogen production, which is the key to large-scale hydrogen utilization. In this study, we developed an off-grid PV-BESS-PEM system for hydrogen and oxygen production. Firstly, the PV, BESS and PEM water electrolysis components were modeled and model validation was performed. Secondly, an energy management strategy (EMS) is proposed to achieve the goal of continuous and stable hydrogen production. Then, a surrogate model between design variables and optimization objectives is established based on machine learning methods, and a multi-objective optimization algorithm is used to drive the optimization process. Finally, a case study of the PV-BESS-PEM system is presented based on solar irradiation data in Lhasa, where there is sufficient solar resources and oxygen demand. The results demonstrate that the EMS can satisfy the purpose of all-day continuous and stable hydrogen production. The number of cells in the electrolyzer and the water temperature have a significant effect on the hydrogen production and the electrolysis efficiency. And the optimal system has increased the hydrogen production by 16.80 % and the electrolysis efficiency of PEM by 12.08 %, respectively. Herein, the methodological framework is expected to provide theoretical guidance for large-scale water electrolysis hydrogen production applications based on solar energy.

1. Introduction

The rapid increase in energy consumption and environmental pollution are affecting the progress and development of today's society [1]. The International Energy Outlook 2021 report predicts that global energy consumption will increase about 50 percent by 2050 [2]. Renewable energy has been developing rapidly for the past years due to its advantages of reliable source, low environmental pollution, low operating cost and easy maintenance. Hydrogen energy is identified as the ideal energy source for the 21st century due to its high calorific value, good combustion performance, zero carbon, convenient transportation and easy storage [3,4]. The generation of electricity from renewable energy sources (RES) to produce hydrogen by electrolysis of water is considered to be the cleanest sustainable technology for hydrogen production. Once the combination of photovoltaic (PV) power generation-battery energy storage-hydrogen production from electrolyzed water is combined, it is possible to reduce the rate of light abandonment and solve the problem of volatility of PV power generation [5,6].

Currently, hydrogen production technologies from electrolyzed water include alkaline water electrolysis (AWE), proton exchange membrane (PEM) electrolysis, and solid oxide electrolysis cells (SOEC) [7]. Among them, AWE and PEM electrolysis for hydrogen production are gradually applied in the commercialization stage. The former hydrogen production technology is mature and low-cost, but the hydrogen production efficiency is low and the performance is poor. The latter is gradually industrialized with high security, high efficiency, high purity of hydrogen and adaptability to power fluctuations from RES [8]. For PV-based electrolyzed water hydrogen production system, there are two modes of connecting the PV array and the electrolyzer, i.e., direct coupling and indirect coupling modes [9]. PV-based hydrogen production indirect coupling system consists of PV array, maximum power point tracking (MPPT) controller, battery energy storage system (BESS),

* Corresponding author.
E-mail address: shaoshq@hust.edu.cn (S. Shao).

https://doi.org/10.1016/j.renene.2024.120483
Received 27 December 2023; Received in revised form 17 March 2024; Accepted 9 April 2024
Available online 10 April 2024

Nomenclature

A_m	Membrane area
C_b	Energy capacity
E_{pro}	Activation energy
F	Faraday constant
$f_{c,f,i}$	Capacity decay factor
I_1	First diode saturation current
I_2	Second diode saturation current
I_{PV}	Solar induced current
j	Current density
k	Boltzmann constant
N	Diode emission coefficient
P	Power
R	Resistance
T	Temperature
V_{PEM}	PEM electrolyzer operating voltage
V_{act}	Activation overpotential
V_{ohm}	Ohmic overpotential
V_{OCV}	Open circuit voltage
V_{conc}	Concentration overpotential
V	Solar cell terminal voltage
y	Predicted value
$\hat{y}$	True value
α	Electron transfer coefficient
β	Overvoltage calculation factors
δ_{mem}	Membrane thickness
σ_{mem}	Membrane conductivity
η	Efficiency

Subscripts

an	Anode
B	Battery
ca	Cathode
H_2	Hydrogen
m	Membrane
n	Number of samples
O_2	Oxygen
p	Parallel
s	Series

Abbreviations

AWE	Alkaline water electrolysis
BESS	Battery energy storage system
EMS	Energy management strategy
GHI	Global horizontal irradiation
GPR	Gaussian process regression
HV	Hypervolume
IGD	Inverse generational distance
MPP	Maximum power point
MPPT	Maximum power point tracking
PEM	Proton exchange membrane
POGCEA	Paired offspring generation based constrained evolutionary algorithm
PV	Photovoltaic
RES	Renewable energy sources
RVGEA	Reference vector guided evolutionary algorithm
SOEC	Solid oxide electrolysis cells
SOC	State of charge
WAS	Weighted average surrogate

DC/DC converter, electrolyzer, etc. In this indirect coupling mode, the electricity generated by the PV array is stored through the BESS and then smoothly released through the DC/DC converter [10]. The direct coupling mode means that the DC power output from the PV array is directly connected to the electrolysis unit, eliminating the need for converters and other equipment [11]. This mode requires that the performance curves of the PV array and the electrolysis unit can be better matched to enable the system to operate efficiently and economically. The integration of RES such as solar and energy storage systems into off-grid PEM water electrolysis plants to achieve continuous and stable hydrogen production faces a number of challenges, and it is a topic of much research interest [12].

Many studies have been reported on the direct coupling systems of PV panels and electrolyzer units. These scholars mainly focus on the analysis and optimization of models to improve the electrolysis efficiency of hydrogen production in direct coupled systems or to reduce the energy loss of the system [13,14]. Yang et al. [15] established a direct-coupled system consisting of PV panels and PEM electrolysis units. By optimizing the hydrogen production efficiency, the optimal serial and parallel numbers of PV cells were obtained. In addition, sensitivity analyses were conducted for some key influencing factor parameters and their effects on the improvement of hydrogen production efficiency were elucidated. Mraoui et al. [16] proposed a hydrogen production model with direct coupling of PV and PEM electrolysis, and simulated the real system by experimentally correcting the numerical and empirical models. Based on this, the direct coupling system was optimized to improve the hydrogen production efficiency. Considering the fluctuating characteristics of solar energy, Cai et al. [17] combined a Markov model with a direct coupled system of PV-PEM electrolyzer to construct reliable operating conditions. Through the genetic algorithm-driven optimization process, the energy transfer loss of the system was drastically reduced with an average coupling efficiency of 98.8 %. To maximize hydrogen production and minimize energy transfer losses, Maroufmashat et al. [18] proposed a multi-objective optimization framework based on an imperialistic competitive algorithm to determine the optimal sizing and operating conditions of the direct coupling system. The results also show that if the optimization is performed with only the hydrogen production as the objective, the hydrogen production will be 2.2 % higher than the optimization result with only the energy transfer loss as the objective, but the energy transfer loss of the system will increase by 68 %. Not coincidentally, Laoun et al. [19] simultaneously optimized the energy loss of the directly coupled system PV-PEM and the hydrogen production rate. A decision-making tool is also proposed for determining the performance of PV-PEM and evaluating the economics of the model. In summary, although the direct coupling model can simplify the composition of the system to a certain extent, eliminating equipment such as converters and reducing the complexity of the system, the performance curves of the PV arrays and electrolysis units are required to be better matched, and thus the system needs to be optimally matched. In addition, the operating parameters of the direct-coupled system need to be precisely adjusted with the dynamic operation of the electrolyzer, which may increase the uncertainty of the system and reduce its reliability.

The indirect coupling method of PV and electrolyzer is currently the most mainstream system coupling mode, which has attracted the attention of many scholars [20–22]. Since the indirectly coupled system of PV and electrolyzer is more complex compared to the directly coupled system, which means that the energy transfer loss of the system is high. How to guarantee the hydrogen production of the model and its economy has become the focus of current research. Several studies have been targeted to propose some optimization models and control strategy. To solve the problem of fluctuating PV power generation, Cao et al. [23] proposed a hydrogen production system consisting of PV panels, BESS and AWE, and designed an energy management strategy (EMS). The

Table 1
Overview of literature on green hydrogen production based on renewable energy.

System description	Coupling mode	Contribution	Hydrogen production/efficiency	Ref
PV-PEM	Direct	The influence of climate conditions was studied.	8.2 kg/year	[14]
PV-PEM	Direct	The expression for the hydrogen production efficiency was derived	7.9 %	[15]
PV-PEM	Direct	The electrolytic efficiency was 53.3 %.	0.45 kg/h	[29]
PV-PEM	Direct	For the direct coupling mode, RMSE value of PV and electrolyzer was about 2 %.	N/A	[16]
PV-PEM	Direct	The influence of PV fluctuation on irreversible attenuation was revealed.	N/A	[17]
PV-AWE	Direct	Utilization of PV resources based on the power-to-hydrogen concept with an electrolysis efficiency of 10.1 %.	10.1 %	[30]
PV-BESS-electrolyzer	Indirect	The electrolytic efficiency was 67.8 % when voltage was 3 V.	1.95 mL/h	[31]
PV-BESS-PEM	Indirect	The influence of different component prices on the cost of hydrogen production was explored.	N/A	[21]
PV-BESS-PEM	Indirect	The technical feasibility of the coupled system of short-term BESS and seasonal hydrogen storage was verified.	170–190 kg/winter	[22]
PV-BESS-AWE	Indirect	An energy management strategy is proposed to reduce energy consumption for hydrogen production.	1.68 Nm^3/8.68 kWh	[23]

designed EMS is verified to ensure that the system operates in a steady state, and the power supplied to the AWE is adjusted every 5 min to ensure its efficient operation. Cecilia et al. [24] proposed a microgrid system consisting of PV-lead-acid battery-electrolyzed water for hydrogen production-load, which incorporates energy management strategies to reduce energy transfer losses and battery size. Two scenarios with and without batteries were investigated [25], and the results showed that the presence of the battery can reduce the size of the electrolytic cell, although it increases the energy transfer loss of the system and the cost. Izadi et al. [26] proposed an adaptable hybrid energy system and performed operational simulations. Optimized parameters for the system configuration were obtained through a genetic algorithm-driven optimization process, which reduced the CO_2 emissions, cost and probability of power supply losses of the system (53.48 tons/year, 1.422 €/hour and 0.4057). Li et al. [27] studied a hydrogen production method coupling photothermal synergistic reaction with PV-electrolyzer, which can fully utilize full-spectrum solar energy. Through the validation of the established model, the hydrogen production efficiency is about 21.05 %. Hybrid hydrogen production systems that combine photothermal synergistic reactions with PV/T electricity generation system can also enhance the efficiency of hydrogen production. Li et al. [28] investigated the hydrogen production efficiency of hybrid systems with different component parameters, operating conditions and component materials. The results show that the decrease of the temperature of the PV panel and the increase of the separation wavelength can improve the efficiency. And when the material of the PV panel is GaAs, the efficiency can reach up to 20.52 %. For the indirect coupling system, which requires some other equipment to work with, the reliability and hydrogen production of the system is improved, which is suitable for large-scale hydrogen production scenarios. However, the investment for the whole system and the maintenance of the equipment are higher, resulting in a mismatch between cost and economic efficiency. Table 1 demonstrates an overview of the literature on hydrogen production from renewable energy.

Currently, few studies have focused on continuous and stable PV-BESS-PEM all-day water electrolysis for hydrogen production. In addition, how to integrate machine learning and intelligent optimization algorithms to optimize the system components and controls to maximize the hydrogen production efficiency and the electrolytic efficiency of the PEM electrolyzer is the key for the future utilization of large-scale PV-BESS-PEM water electrolysis for hydrogen production [32]. In this study, we propose an off-grid PV-BESS-PEM electrolytic water hydrogen production system, and design an EMS to meet the system's demand for continuous and stable hydrogen production around the clock. A case study was also conducted using solar irradiation data from Lhasa, China, to optimize the hydrogen production and the electrolytic efficiency of the PEM by combining the surrogate model with the intelligent optimization algorithm. The optimized hydrogen production of the system

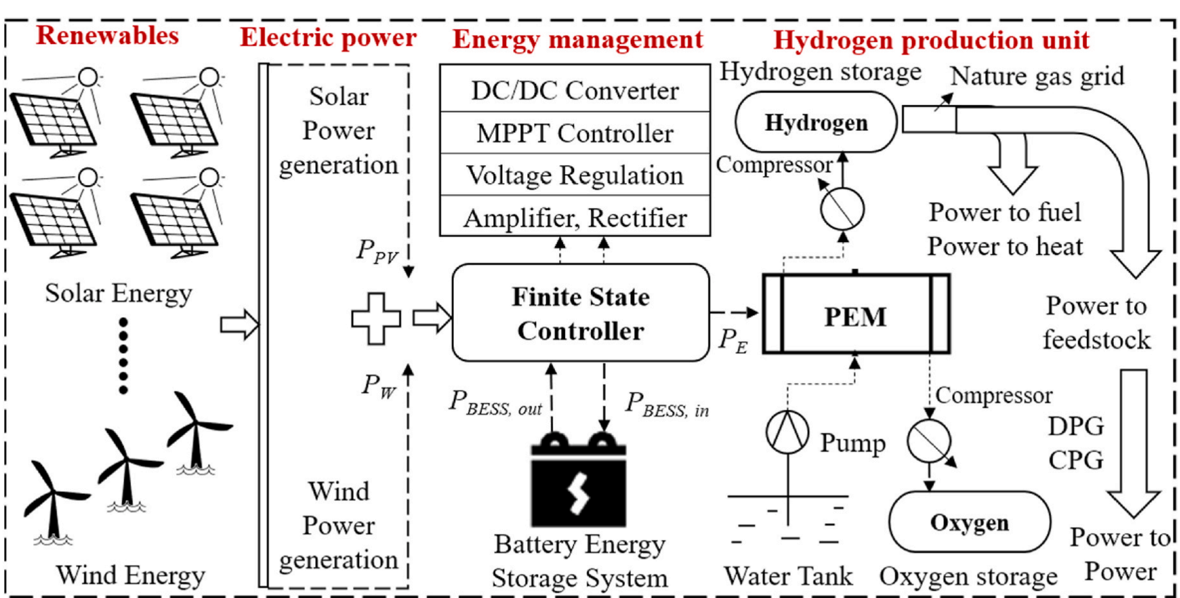

Fig. 1. Schematic of RES-energy storage-electrolysis-hydrogen and oxygen utilization hybrid energy system for power, hydrogen and oxygen generation.

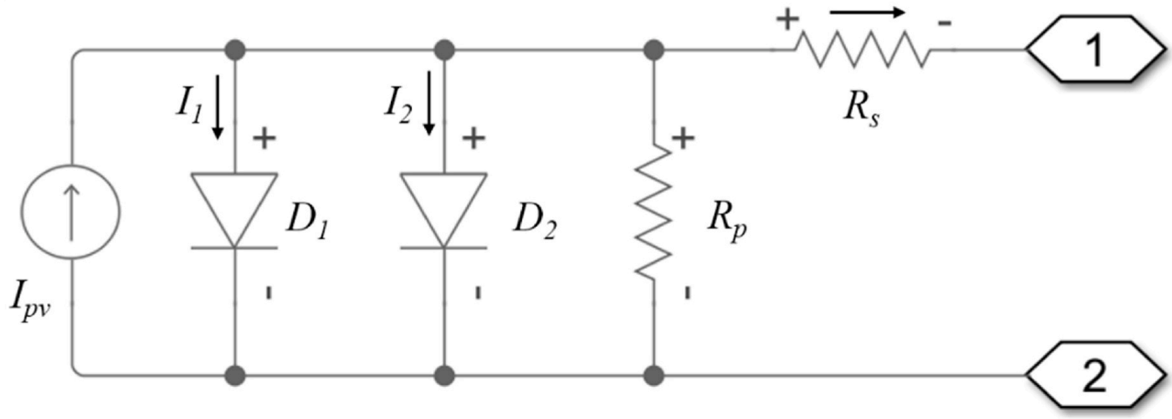

Fig. 2. Solar cell equivalent circuit diagram.

can satisfy the heating of an office building, which provides a new idea for the large-scale application of hydrogen production from electrolytic water.

The reminder of this paper is organized as follows. The off-grid PV-BESS-PEM electrolytic water for hydrogen production system and modeling of the PV, BESS, and PEM electrolyzer are described in Section 2. Section 3 introduces the EMS and carries out surrogate model establishment, and optimization. Section 4 presents a case study, where a sensitivity analysis of the system is performed and the optimization results are compared. Section 5 summarizes the main conclusion.

2. System description and components modeling

2.1. System description

Fig. 1 illustrates a schematic of a hybrid energy system for power generation-energy storage-electrolysis of water for hydrogen and oxygen production-hydrogen and oxygen utilization. In this study, the hydrogen production and electrolytic efficiency of the hybrid energy system based on PV-BESS-PEM electrolysis of water for hydrogen and oxygen production is mainly evaluated. The hybrid energy system consists of three components: PV panels, BESS, and PEM electrolyzer for oxygen and hydrogen production. And the optimization of the PV hydrogen-oxygen cogeneration energy system is performed [33]. The detailed modeling process for each component will be introduced subsequently. It is worth noting that the developed model is programmed in MATLAB software and computed by numerical simulation in Simulink.

2.2. PV array model

The PV array module consists of solar cells connected in series and parallel, and output varies according to the increase or decrease in the incident light intensity [34]. In this study, the equivalent circuit model of solar cells is chosen as the base model for simulation calculations, and the equivalent circuit model of a single solar cell is shown in Fig. 2. The model contains a current source, two exponential diodes, a parallel resistor and a series resistor.

When the irradiance (light intensity) on the cell is I_r, the output current I_{out} of a single solar cell can be calculated as follows [29]:

$$I_{out} = I_{pv} - I_1 \cdot \left(e^{\frac{(V+I_{out}\cdot R_s)}{N_1 \cdot kT/q}} - 1\right) - I_2 \cdot \left(e^{\frac{(V+I_{out}\cdot R_s)}{N_2 \cdot kT/q}} - 1\right) - \frac{(V + I_{out}\cdot R_s)}{R_p} \quad (1)$$

$$I_{pv} = I_{pv0}\cdot(I_r / I_{r0}) \quad (2)$$

where I_{pv} is the solar induced current, I_1 and I_2 are the saturation current of the first and second diode respectively, V is the solar cell terminal voltage, $k = 1.380649 \times 10^{-23}$ J/K, T is the solar cell temperature, q is the charge, N_1 and N_2 are the diode emission coefficient of the first and second diode respectively, R_s is the series resistance, and R_p is the parallel resistance.

The saturation currents I_1 and I_2 of the first and second diodes are affected by the temperature. According to the solar cell temperature T, I_1 and I_2 can be calculated as follows:

$$I_1(T) = I_1 \times \left(\frac{T}{T_{means}}\right)^{(TXI1/N_1)} \times e^{\left(EG*\left(\frac{T}{T_{means}}-1\right) \Big/ (N_1 * V)\right)} \quad (3)$$

$$I_2(T) = I_2 \times \left(\frac{T}{T_{means}}\right)^{(TXI2/N_2)} \times e^{\left(EG*\left(\frac{T}{T_{means}}-1\right) \Big/ (N_2 * V)\right)} \quad (4)$$

Table 2
The specifications of PV component.

Parameter	Value	Unit
Short circuit current	3.80	A
Open circuit voltage	0.59	V
Quality factor, N_1	1.40	N/A
Quality factor, N_2	2.00	N/A
Energy gap	1.40	eV
Measurement temperature	25	°C

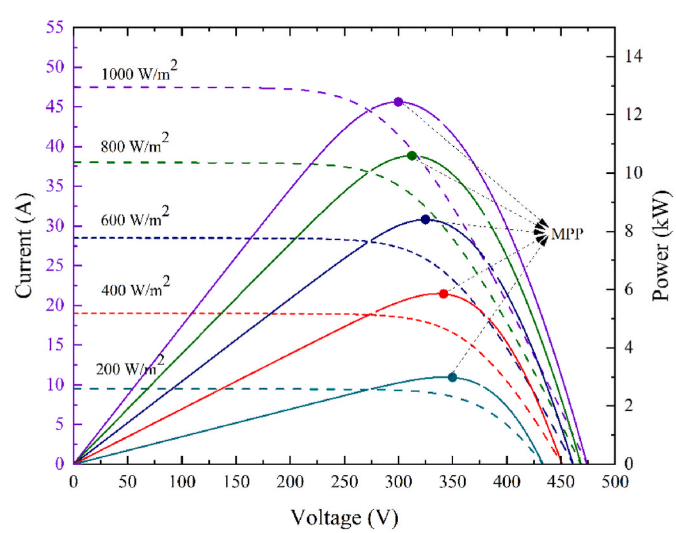

Fig. 3. Current–voltage and power-voltage of the PV array at different irradiance.

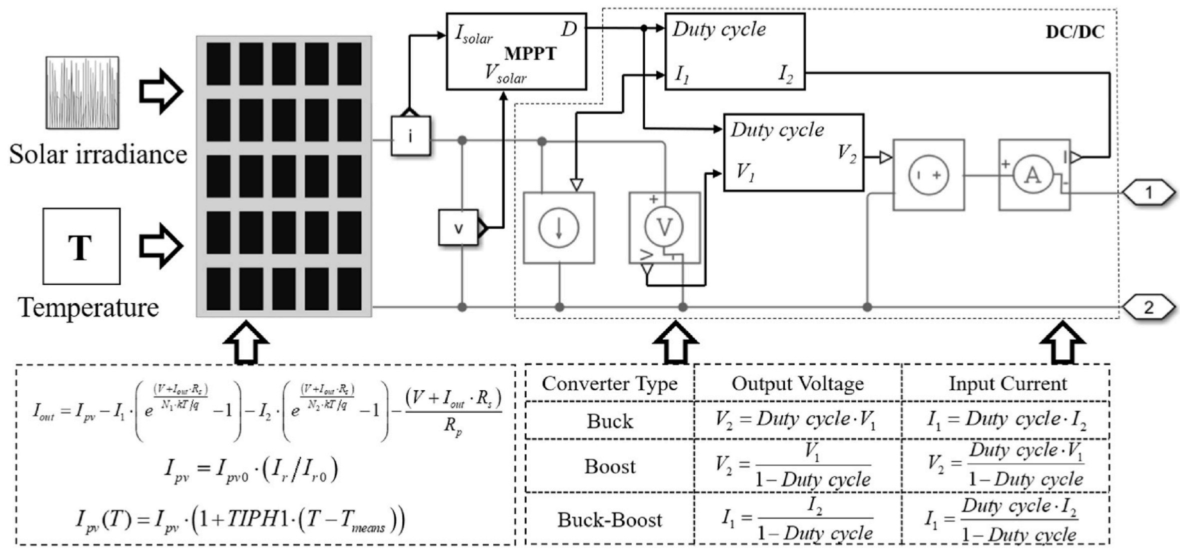

Fig. 4. PV panel, Control algorithm and related model for PV power generation.

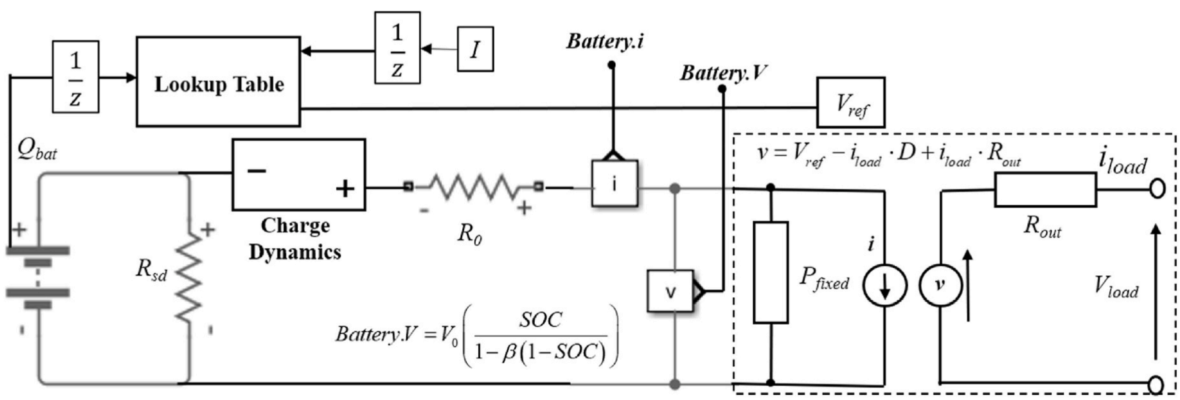

Fig. 5. Modeling of battery energy storage system.

where T_{means} is the measured value of temperature, $TXI1$ is the temperature exponent value of I_1, $TXI2$ is the temperature exponent value of I_2. The component parameters of PV are presented in Table 2.

The output current-voltage-power characteristics of the PV array model at different irradiance levels when the temperature is 298.15 K are demonstrated in Fig. 3. From Fig. 3, it can be seen that the power and current of the PV array varies with irradiance. When the irradiance and temperature are constant, the output power first increases and then decreases with the increase of voltage, and the curve has only one extreme value, that is, the maximum power point (MPP). Therefore, the most ideal state is to make the PV panels work at near the MPP all the time by adjusting the voltage of the PV array. However, duo to the intermittent nature of irradiance causes the MPP to change over time. If it is not possible to track the MPP, it will cause serious power loss and energy waste.

To address the above problems, a maximum power point tracking (MTTP) algorithm is adopted in this study to maximize the PV power generation. The PV module is electrically connected to the load (electrolyzed water unit) through a DC-DC converter [35]. As shown in Fig. 4, the MTTP algorithm is adopted to the PV system to coordinate the work of the solar cells, the battery storage system and the water electrolysis unit. The voltage and current changes of the PV array are constantly detected by the MPPT algorithm, and the duty cycle of the DC-DC converter is adjusted according to the changes. Fig. 4 also shows that the duty cycle determines whether the DC-DC converter is a buck converter, a boost converter, or a buck-boost converter to match the load power.

2.3. BESS modeling

BESS has the advantages of relatively mature technology, safety and reliability, low noise, large capacity, strong environmental adaptability, easy to install and so on, and has been increasingly popularized and applied [36]. This study refers to the model developed by Ibáñez-Rioja et al. [21] and builds on it to develop the BESS model for this paper. The BESS is electrically connected to the DC bus via a DC/DC converter as shown in Fig. 5. The charging and discharging mode of the battery is controlled by an EMS.

The relationship between voltage and state of charge (SOC) can be calculated as follows:

$$V_B = V_0 \left(\frac{SOC}{1 - \beta(1 - SOC)}\right) \tag{5}$$

where V_0 is the rated voltage when fully charged and β is a constant.

The efficiency of the BESS is assumed to be constant during charging and discharging and $\eta = 0.92$, round trip efficiency is 90 % [37]. The charging and discharging states of the BESS are controlled by the proposed EMS and there are charging and discharging modes. The power P of the BESS in charging mode is defined as [21,38]:

Table 3
The specifications of BESS component.

Parameter	Value	Parameter	Value
Nominal voltage	240 V	Cell capacity	1e6 hr*A
Current directionality	Disabled	Self-discharge	disabled
Internal resistance	2 Ohm	Voltage *V1* when charge is *V1*	210 V
Battery charge capacity	Finite	Charge *AH1* when no-load voltage is *V1*	5e5 hr*A

$$P_{c,i} = \min\left\{\eta P_{c,input,i}, (1 - SOC_{i-1})(1 - f_{c,f,i})C_b / \Delta t, \eta P_{c,\lim,i}\right\} \tag{6}$$

where η is the efficiency of the BESS, $P_{c,input,i}$ is the input power, $P_{c,lim,i}$ is the power limit, C_b is the energy capacity, $f_{c,f,i}$ is the capacity decay caused by degradation.

The degradation of the battery capacity leads to a power drop, which affects its SOC. In charging mode, the calculation formula of SOC after charging is as follows:

$$SOC_i = SOC_{i-1} + \frac{P_{c,i}\Delta t}{C_b} \tag{7}$$

Taking the minimum value of power demand, available charge, and the power limit as the battery power in the battery discharge mode, which can be expressed as follows:

$$P_{d,i} = -\min\left\{P_{d,demand,i}, \eta SOC_{i-1} C_b / \Delta t, P_{d,\lim,i}\right\} \tag{8}$$

where $P_{d,demand,i}$ is the power demand, C_b is the energy capacity and $P_{d,lim,i}$ is the power limit.

For the BESS, current flows into the battery in charging mode and flows out of the battery in discharging mode. Therefore, it is stipulated that the battery power in charging mode is positive and the battery power in discharging mode is negatively signaled. In discharging mode, the calculation formula of SOC after discharging is as follows:

$$SOC_i = SOC_{i-1} + \frac{P_{d,i}\Delta t}{\eta C_b} \tag{9}$$

The component parameters of BESS are presented in Table 3.

2.4. PEM electrolyzer modeling

PEM electrolyzer uses solid PEM as electrolyte and pure water as reactant, including polymer membrane, catalyst, gas diffusion layer and bipolar plate [39]. The anodic and cathodic reactions of water electrolysis in the PEM electrolyzer are presented as:

$$Anodic\ half-reaction: H_2O \rightarrow 2H^+ + 2e^{-1} + 0.5O_2 \tag{10}$$

$$Cathodic\ half-reaction: 2H^+ + 2e^{-1} \rightarrow H_2 \tag{11}$$

The operating voltage of PEM electrolyzer is calculated as follows:

$$V_{PEM} = V_{OCV} + V_{act} + V_{ohm} + V_{conc} \tag{12}$$

where V_{OCV} is the open circuit voltage, V_{act} is the activation overpotential, V_{ohm} is the ohmic overpotential and V_{conc} is the concentration overpotential.

V_{OCV} is the basic voltage for the water electrolysis reaction to proceed and is equal to the sum of the reversible potentials of each half-reaction at the two electrodes. V_{OCV} can be expressed by the reversible voltage of the water electrolysis reaction and the Nernst equation as follows [40]:

$$V_{OCV} = V_{rev} + \frac{RT}{2F}\ln\left(p_{H_2}\left(\frac{p_{O^2}}{1\times10^5}\right)^{\frac{1}{2}} \Big/ p_{H_2O}\right) \tag{13}$$

where R is the ideal gas constant, T is the operating temperature of the electrolyzer, p_{H_2} and p_{O_2} denote the partial pressures of hydrogen and oxygen, p_{H_2O} denotes the partial pressure of water. V_{rev} denotes the reversible voltage of the water electrolysis reaction, which can be found from the change in Gibbs free energy ($T = 298.15$ K, $p = 100$ kPa). p_{H_2O} can be given by the following empirical formula [41]:

$$p_{H_2O} = 618.87\exp\left[17.2694\cdot\left((T-273.15)/_{T-34.85}\right)\right] \tag{14}$$

The activation overvoltage V_{act} is generated by electrochemical polarization and represents the phenomenon in which the retardation of the electrolytic reaction proceeds leading to a deviation of the electrode potential from the reversible voltage. The activation overvoltage V_{act} can be represented by an expression based on the Butler-Volmer equation as follows [42]:

$$V_{act} = \frac{RT}{2\alpha_{an}F} a\sinh\left(\frac{i}{2i_{0,an}}\right) + \frac{RT}{2\alpha_{ca}F} a\sinh\left(\frac{i}{2i_{0,ca}}\right) \tag{15}$$

where α_{an} is the anode electron transfer coefficient, α_{ca} is the cathode electron transfer coefficient, $i_{0,an}$ and $i_{0,ca}$ are the exchange current density of the anode and cathode of the electrolyzer, respectively. $i_{0,an}$ and $i_{0,ca}$ are affected by the electrode material, surface state and other factors.

Ohmic polarization is produced primarily by the resistance of the PEM to proton transfer, but also by the impediment of current by components such as bipolar plates and electrodes. The calculation of ohmic overvoltage follows Ohm's law, expressed as follows:

$$V_{ohm} = \left(R_{elec} + \frac{\delta_{mem}}{\sigma_{mem}}\right)\cdot i \tag{16}$$

where R_{elec} represents the resistance generated by components such as electrodes, δ_{mem} is the thickness of the PEM, and σ_{mem} is the conductivity of the PEM. The relationship between membrane conductivity σ_{mem} and temperature change can be obtained by the Arrhenius formula:

$$\sigma_{mem} = \sigma_{mem,std}\cdot\exp\left[-\frac{E_{pro}}{R}\left(\frac{1}{T} - \frac{1}{T_{std}}\right)\right] \tag{17}$$

where $\sigma_{mem,std}$ denotes the conductivity of the PEM at standard conditions and E_{pro} denotes the activation energy required for proton transfer in the PEM.

The concentration overpotential is mainly generated by the mass transfer limit of the reactants in the electrode, and at higher current densities, the inability of the produced gases to be removed from the electrode in time also generates additional polarization. By introducing the limiting diffusion current density i_d, the concentration overpotential V_{conc} can be expressed as:

$$V_{conc} = i\cdot\left(\beta_1\frac{i}{i_d}\right)^{\beta_2} \tag{18}$$

where β_1 and β_2 are overvoltage calculation factors. The component parameters of PEM are presented in Table 4.

Table 4
The specifications of PEM component.

Parameter	Value	Parameter	Value
Plate.*X*	50 cm	Membrane area	30,000 cm^2
Plate.*Y*	100 cm	Membrane thickness	2 cm
Plate.*Z*	100 cm	Resistance	0.25 Ohm
Number of electrode pairs	3	Heating resistor	25 Ohm
Number of cells	70	H_2/O_2 tank volume	100*100*100 cm^3
Temperature of H_2 tank	273.15 K	Temperature of O_2 tank	273.15 K

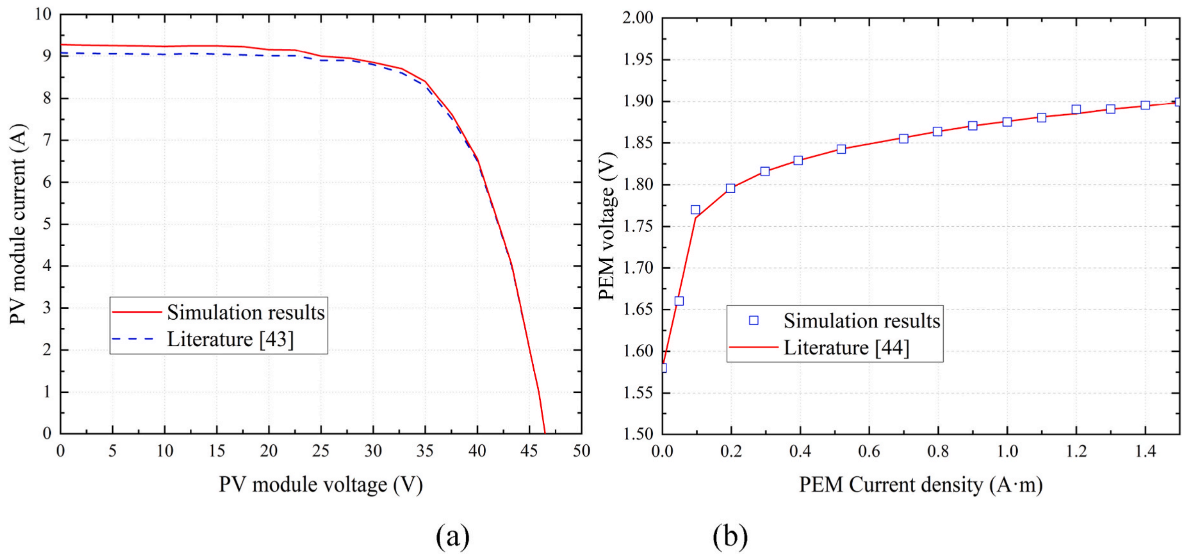

Fig. 6. Model validation of PV (a) and PEM electrolyzer (b).

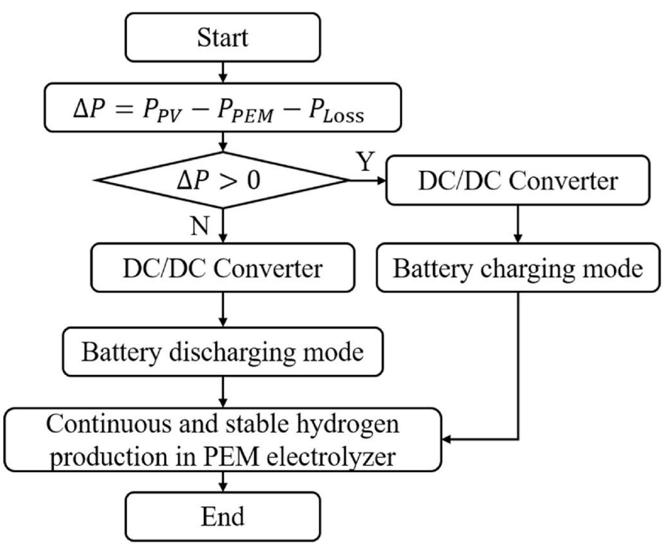

Fig. 7. The process overview of EMS.

2.5. *Model validation*

To guarantee the accuracy of the establishment model, model validation is necessary, which is also a basic for subsequent work. In this study, based on the above modeling process we selected literature [43, 44] to validate the PV model and PEM model, which is also the main components of the system. The main focus is to validate the data in the literature through the modeling approach of this paper, and the actual model in this paper is different from the literature mentioned above. The model validation results are shown in Fig. 6. The results demonstrate that the maximum error between the calculation results and the results in the literature is 2.2 % for the PV model, and the maximum error between the calculation results and the results in the literature is not more than 1 % for the PEM electrolyzer model, which proves the accuracy of the model developed in this study.

3. Solution methodology

3.1. *Energy management strategy*

To guarantee that the system can produce hydrogen continuously and stably around the clock, an EMS is adopted in this paper, as shown in Fig. 7. During the operation of the system, i) when the solar radiation is stronger (midday hours), the power generated by the PV panel is large, and the PV panel power is greater than the sum of the power required by the PEM to electrolyze the water and the power loss, i.e., P_{PV}-P_{PEM}-P_{LOSS} is greater than zero. Then the EMS controls the battery to be in charging mode thus storing the remaining energy. ii) When the solar radiation is weak (at night or early morning hours), the power generated by the PV panels is small, and the power of the PV panels is not enough to support the power required for PEM electrolysis of water and the power loss, i.e., P_{PV}-P_{PEM}-P_{LOSS} is less than zero. Then the battery is in the discharging mode to ensure the stable operation of electrolysis cells.

The solar radiation from 12:00 to 18:00 on a certain day is used as an input to simulate the operating results of the system based on Matlab/Simulink to verify the correctness of the EMS. As shown in Fig. 8, the solar irradiance is less than 400 W/m^2 near 12 o'clock, and the PV power is slightly larger than PEM electrolyzer. However, due to the energy transfer loss of the component, it is practically impossible to satisfy the PEM electrolyzer works continuously. So, the BESS is needed to output power to the outside to maintain the operation of the electrolyzer. At this time, the battery is in the discharge model, and the SOC of the battery is slowly decreasing. When the solar irradiance increases, the PV panel power gradually increases and is greater than the sum of the energy transfer loss and the power of PEM electrolyzed water, at which time the battery does not need to provide additional energy to the outside to maintain the stable operation of the PEM electrolyzed water. In this case, the battery switches to charging mode to store the excess power of the PV. In addition, Fig. 8 (d) demonstrate that no matter how the solar irradiance varies with time, the voltage of the DC bus in the system always remains stable, which verifies the accuracy of the EMS.

3.2. *Optimization with evolutionary algorithms*

3.2.1. *Data-driven surrogate model*

The number of cells in the electrolyzer (x_1), the water temperature (x_2), the membrane area influencing factors *MEMBRANE.Y* and *MEMBRANE.Z* (x_3, x_4), and the membrane thickness (x_5) were selected as the design variables. The hydrogen production of the PV-BESS-PEM water

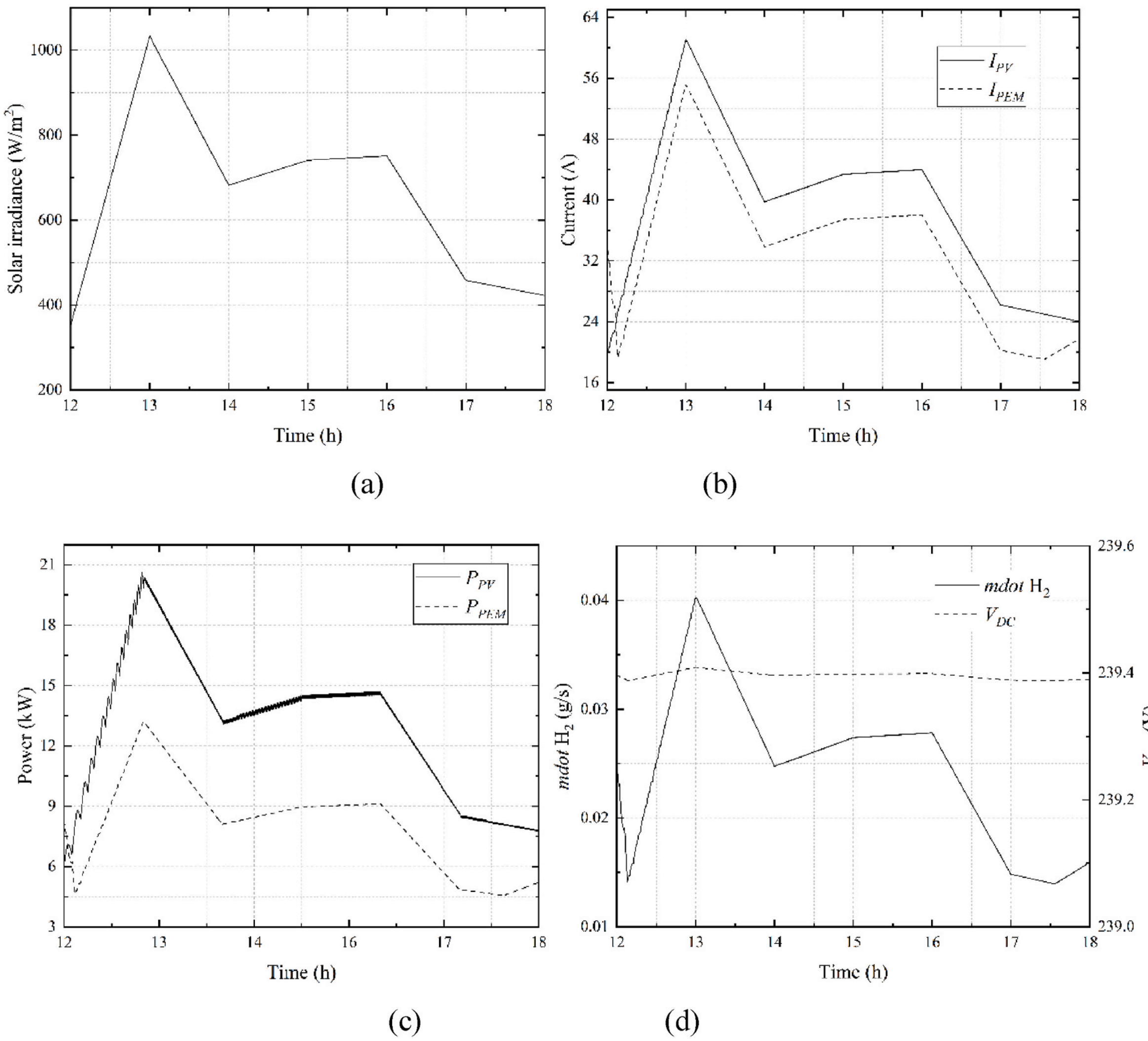

Fig. 8. Solar irradiance (a) and Simulink calculation results of current (b) and power (c) of PV and PEM, hydrogen production rate (d).

Table 5
Results of orthogonal experiments.

Number	x_1	x_2	x_3	x_4	x_5
1	1	1	1	1	1
2	1	2	3	4	1
3	1	3	5	7	2
…	…	…	…	…	…
20	3	1	2	5	1
…	…	…	…	…	…
47	7	5	3	1	2
48	7	3	5	4	3
49	7	4	7	7	3

electrolysis system and the electrolytic efficiency of the PEM were the optimization objectives. The electrolysis efficiency of PEM electrolyzed water is calculated as follows.

$$\eta_{PEM} = \frac{jA_m \mathrm{LHV}_{\mathrm{H_2}}}{2\mathrm{FP}} \tag{19}$$

where j is the current density, unit: A/cm^2, A_m is the membrane area, unit: cm^2, LHV_{H_2} is the low level calorific value of hydrogen which is 10,790 kJ/Nm^3, F is 96,487 C/mol, P is the PEM electrolyzer power, unit: W.

Before building a data-driven surrogate model, it is necessary to obtain the values of the design variables with the optimization objective. Orthogonal experiment was used for experimental design, which was close to the standard orthogonal table $L49$ (7^8). For orthogonal table $L49$ (7^8), *8* represents the number of factors, *7* represents the number of levels, and *49* represents the number of experiments. The standard L49 (7^8) orthogonal table did not satisfy the expected results, so the pseudo-horizontal method was used to correct the standard $L49$ (7^8) orthogonal table. Quasi-level method has two ways of correction: i) If there are redundant factors, directly delete. ii) If a factor has more levels than expected (e.g., factor 1 has five levels, but only four are needed), simply replace the fifth level with one or more of the other four levels. Table 5 shows the partial orthogonal experimental results of this paper.

The surrogate modeling approach shows significant advantages in solving nonlinear modeling problems [45]. Based on the results of the above experimental design, sample data were computed and obtained. Gaussian process regression (GPR) and weighted average surrogate (WAS) were used to construct the surrogate model between design variables and optimization objectives, respectively. The model training results for both optimization objectives were obtained by training the model as shown in Fig. 9. Overall the predicted values of the model are closer to the true values.

RMSE and R-square (introduced in Equations (20) and (21)) were used to specifically present the accuracy evaluation of the model. The results show that for first optimization objective, the R-square value of GPR reaches 0.98, and for the second objective, WAS has an R-square value of 0.99, which is a more desirable result. Therefore, in this paper,

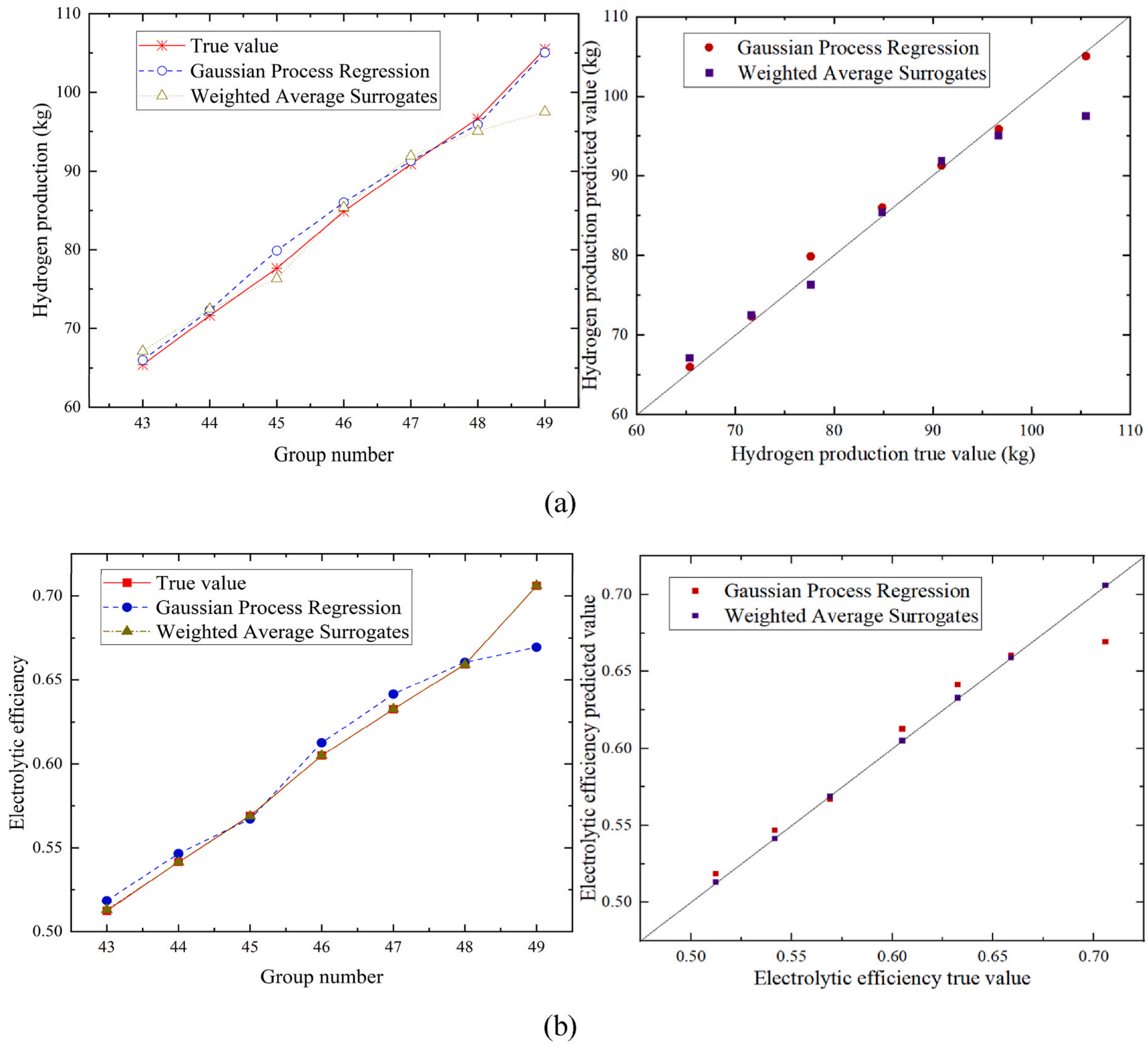

Fig. 9. Model training results of three objectives under different methods. (a) Hydrogen production (b) Electrolytic efficiency.

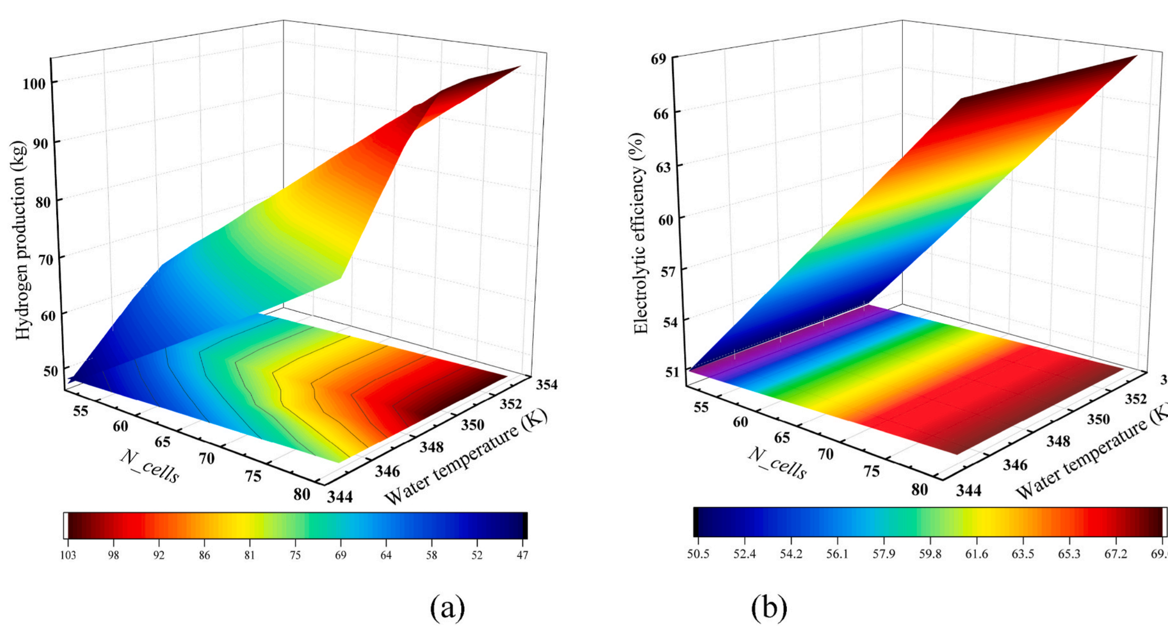

Fig. 10. Variation of hydrogen production (a) and electrolysis efficiency (b) with the number of cells in the electrolyzer and water temperature.

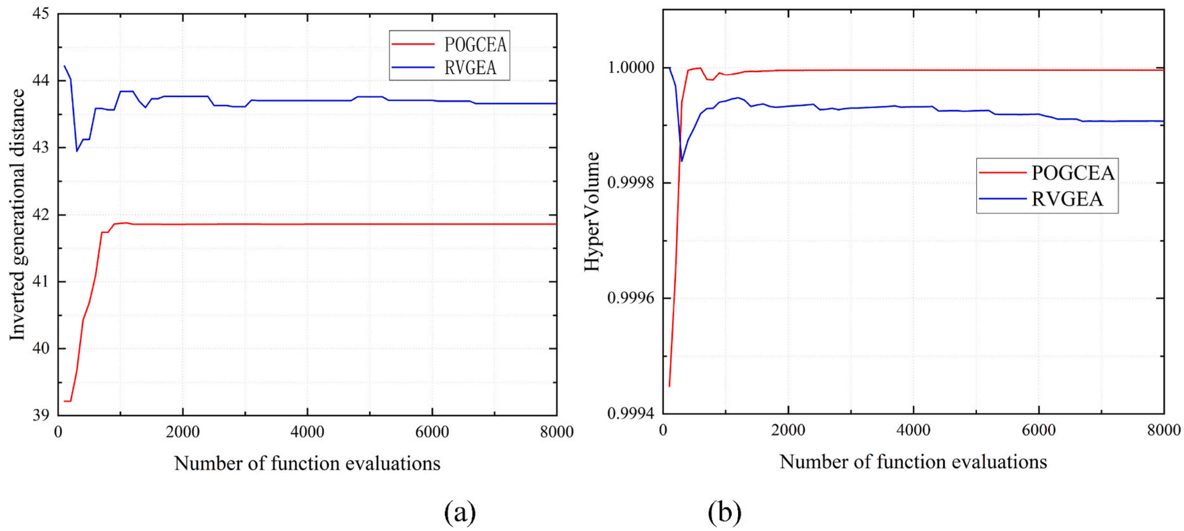

Fig. 11. Result display of the values of IGD-metric (a) and HV-metric (b).

GPR is used to establish the surrogate model for first optimization objective and WAS is used to establish the surrogate model for the second optimization objective.

$$RMSE(y, \hat{y}) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{20}$$

$$R - square(y, \hat{y}) = \left|1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - mean(\hat{y}_i))^2}\right| \tag{21}$$

where y is the predicted value of the model, $\hat{y}$ is the true value, n is the number of samples.

Fig. 10 shows the response surface of the model for different electrolyzer cell numbers and water temperatures. Fig. 10 (a) shows the influence of x_1 and x_2 on hydrogen production. Fig. 10 (b) demonstrates the influence of x_1 and x_2 on electrolysis efficiency. Fig. 10 reveals that the hydrogen production and electrolytic efficiency gradually increases with the increase of x_1, and with the increase of water temperature the hydrogen production first increases and then tends to stabilize.

3.2.2. Optimization model and algorithm

The optimization model in this paper has two optimization objectives and five optimization variables, which can be expressed as equation (22). It is worth noting that x_1 can only take integers during the optimization process, while x_2, x_3, x_4, and x_5 are all consecutive, making this a hybrid optimization problem. In this study, a paired offspring generation based constrained evolutionary algorithm (POGCEA) [46] is used to drive the optimization process. The optimization results are compared with the optimization results using the reference vector guided evolutionary algorithm (RVGEA) [47].

$$\begin{cases} \min F(x) = \left(\max f_{Hydrogen\ production}, \max f_{Electrolytic\ \text{efficiency}}\right) \\ \quad x = (x_1, x_2, x_3, x_4, x_5) \\ \quad f_{Hydrogen\ production} = f_{GP}(x) \\ \quad f_{Electrolytic\ \text{efficiency}} = f_{WAS}(x) \\ s.t.\ \ 50 \le x_1 \le 80, 341.15 \le x_2 \le 350.15, \\ \quad 100 \le x_3 \le 130, 100 \le x_4 \le 130, \\ \quad 0.5 \le x_5 \le 2, x_1 \in Z. \end{cases} \tag{22}$$

The inverse generational distance (IGD) metric and the hypervolume (HV) metric are often used to evaluate the superiority of the two algorithms in solving this problem. The values of IGD metric and HV metric for different number of function evaluations are shown in Fig. 11. The smaller the IGD value and the larger the HV value prove the better the performance of the algorithm. From Fig. 11, it can be seen that the IGD value of POGCEA is smaller and HV value is larger than RVGEA and

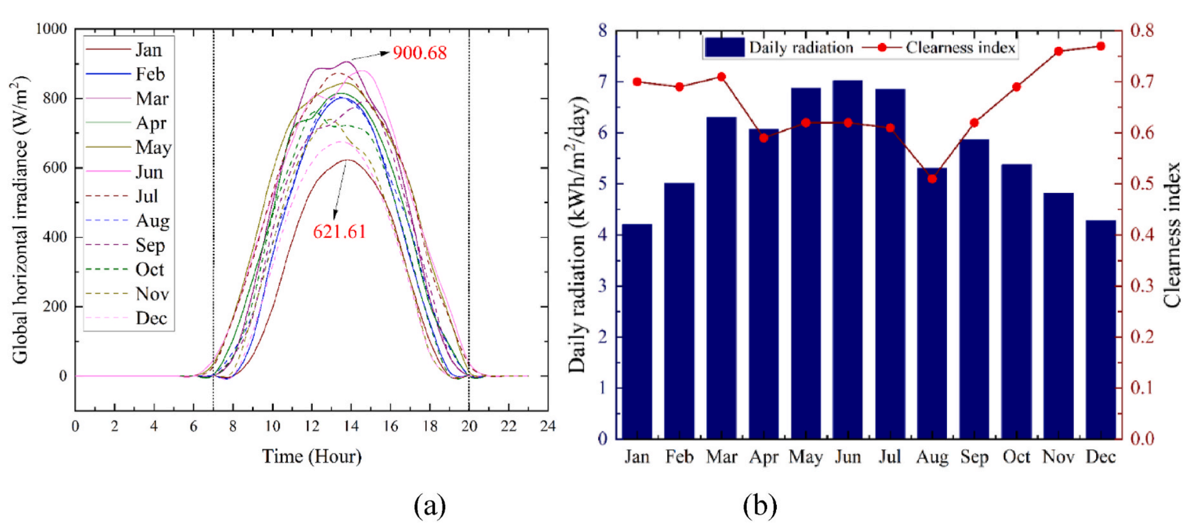

Fig. 12. (a) Solar GHI daily profile and (b) Daily radiation and clearness index at each month.

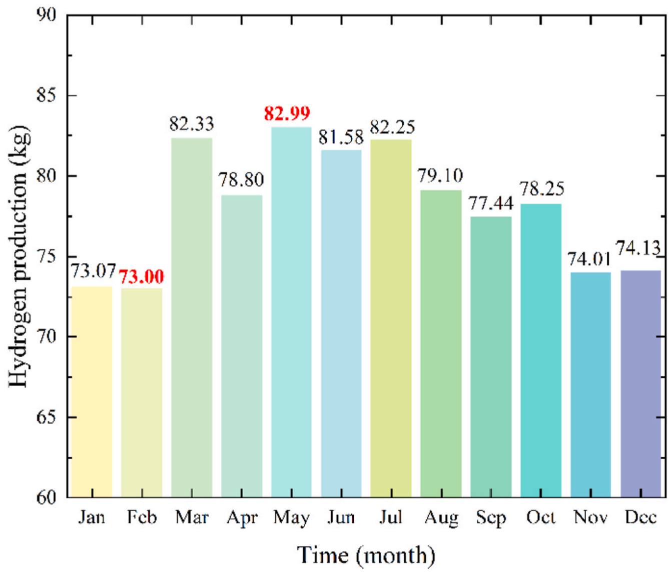

Fig. 13. Monthly hydrogen production of the PV-BESS-PEM system.

POGCEA converges faster. The results show that POGCEA outperforms RVGEA in solving this problem.

4. Result and discussion

4.1. Case study

Herein, the numerical simulation and validation of the established PV-BESS-PEM water electrolysis system are carried out, and the optimization results are discussed. Lhasa (latitude and longitude: 29.6526N°, 91.1378E°), located in Xizang, China, is rich in solar radiation and is the preferred address for PV-based power generation. Global horizontal irradiation (GHI) is defined as the total solar radiation received on the horizontal surface, which includes both direct normal irradiance and diffuse horizontal irradiance.

Fig. 12 (a) shows the solar GHI daily profile. It can be seen from Fig. 12 (a) that the sunrise-sunset time is the longest in June of the whole year, that is, the solar radiation time is the longest. For the Lhasa area, the daily cycle of solar GHI is between about 7:00 a.m. and 20:00 p.m., and the maximum GHI is about at 12:00–15:00 p.m. In addition, the maximum value of daily solar irradiation occurs in March and the minimum in January, with maximum and minimum values of 900.68 W/m^2 and 621.61 W/m^2, respectively. Fig. 12 (b) shows the daily radiation and clearness index at each month. Monthly clearness index is the ratio of the total monthly mean daily solar radiation incident on the horizontal surface to the astronomical radiation (i.e., the monthly mean daily solar radiation reaching the upper boundary of the atmosphere) [48]. Fig. 12 (b) reveals that the bright period from May to July in Lhasa region, and its daily radiations is 6.88, 7.03, and 6.85, respectively. The cloudy period was from November to January of the following year, and its daily radiation was 4.83 kWh/m^2/day, 4.28 kWh/m^2/day, and 4.21 kWh/m^2/day, respectively. In addition, the monthly clearness index for the region fluctuated within a range of 0.51 (Aug) to 0.77 (Dec).

Based on the established PV-BESS-PEM electrolytic water hydrogen production system, the system was simulated with reference to the solar radiation data in Lhasa. From Fig. 13, we know that the highest hydrogen production of 82.99 kg was recorded in the month of May throughout the year and the lowest hydrogen production of 73.00 kg was recorded in the month of Feb. Although January's daily radiation was smaller than February's, the fact that there were three fewer days in February than in January caused February's hydrogen production to be slightly lower than that of January. Similarly, the highest hydrogen

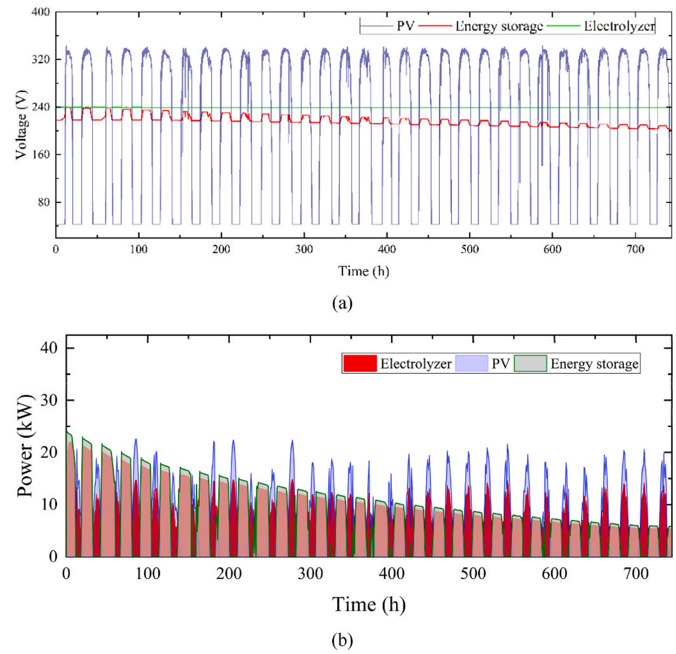

Fig. 14. Operating voltage (a) and power (b) of PV, BESS and PEM electrolyzer.

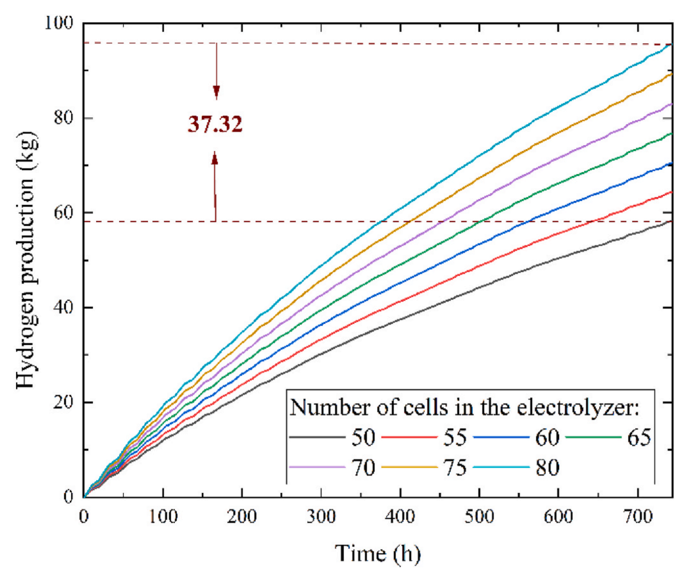

Fig. 15. Effect of electrolytic cell number on hydrogen production in PV-BESS-PEM water electrolysis system.

production in May was caused by this reason. In addition, it is not difficult to notice in Fig. 13 that the hydrogen production in the light period is higher than that in the dark period.

The performance of each component of the PV-BESS-PEM system was investigated in May, the month with the highest hydrogen production. Fig. 14 shows the voltage and power of the PV, BESS and PEM electrolyzer during the electrolysis of water for hydrogen production. From Fig. 14 (a), it can be seen that the voltage generated by the PV panel fluctuates a lot due to the variation of GHI, but the voltage of the electrolyzed water, i.e., the voltage of the DC bus, is always stabilized due to the EMS and the BESS. At the end of the month, the voltage of the battery tends to decrease, which indicates that the battery is generally discharged throughout the electrolysis process. From Fig. 14 (b), it is not difficult to see that when the power generated by the PV is small, the BESS outputs energy to ensure the continuous and stable operation of

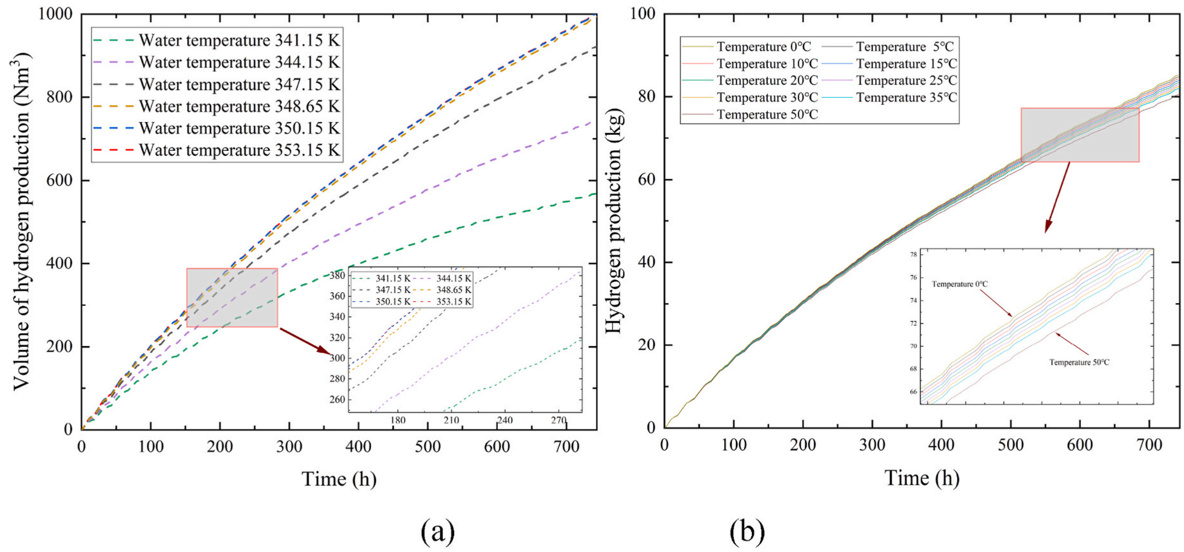

Fig. 16. Effect of water temperature (a) and PV temperature (b) on hydrogen production in PV-BESS-PEM water electrolysis system.

the PEM electrolysis and its output power is slightly larger than the power of the PEM electrolysis (due to the energy transfer loss). At the end of the month, the output power of the battery decreases as it is overall in a discharged state. This also validates the correctness of the proposed EMS.

4.2. Sensitivity analysis

The parameter sensitivity analysis of the system is carried out to study the influence of different electrolytic cell number, water temperature, membrane area and membrane thickness. The sensitivity analysis of the PV-BESS-PEM system was able to adequately analyze the factors limiting the improvement of the hydrogen production of the system and show the potential to improve the hydrogen production of the studied system.

4.2.1. Effect of electrolytic cell number

Fig. 15 shows the influence of hydrogen production of the system with different number of electrolytic cells, taking May as an example. Fig. 15 demonstrates that when the number of cells in the electrolyzer increases, the hydrogen production also increases, and there is a positive correlation between the electrolytic cell number and hydrogen production. The minimum hydrogen production of the system is 58.31 kg when the number of electrolytic cells is 50, and the maximum hydrogen production of the system is 95.63 kg when the number of electrolytic cells is 80. Compared with the number of 50 cells in the electrolyzer, the hydrogen production of the PV-BESS-PEM electrolysis water system increased by 37.32 kg and 64.00 % when the number of 80 cells in the electrolyzer.

4.2.2. Effect of water/PV temperature

The influence of the water temperature in the PEM electrolyzer and the PV panels temperature on the hydrogen production of the PV-BESS-PEM water electrolyzer system is shown in Fig. 16. Fig. 16 demonstrates that as the water temperature increases, the hydrogen production of the system generally shows an increasing trend, but the amount of change gradually decreases. For example, the change in water temperature from 341.15 K to 344.15 K increases the hydrogen volume production from 567.82 Nm^3 to 744.92 Nm^3, which is an increase of 31.19 %, while the change in water temperature from 347.15 K to 353.15 K increases the

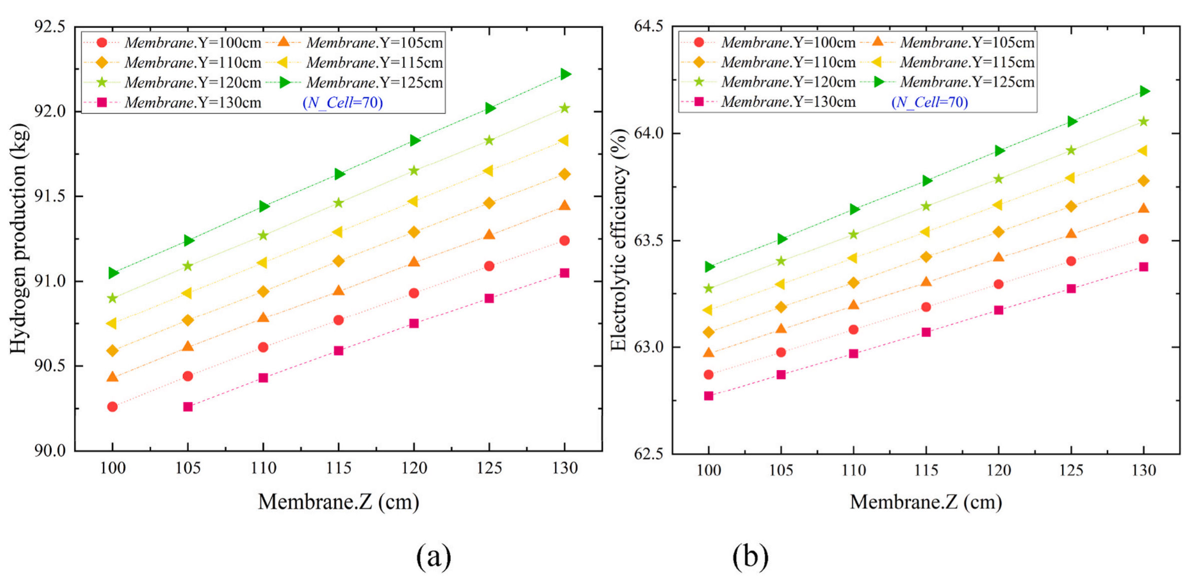

Fig. 17. Effect of membrane area on hydrogen production (a) and electrolytic efficiency of electrolyzer (b) in PV-BESS-PEM water electrolysis system.

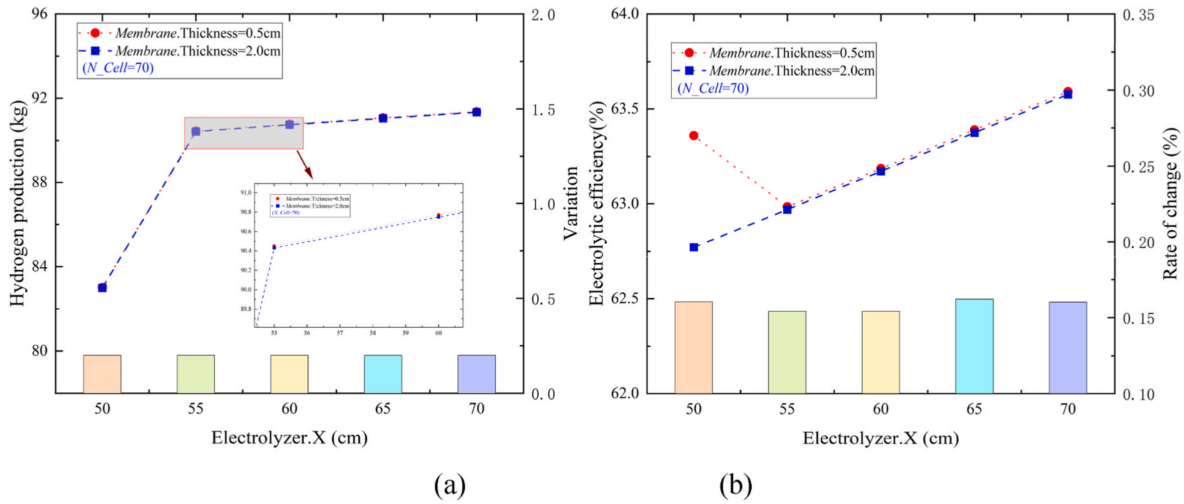

Fig. 18. Effect of membrane thickness on hydrogen production (a) and electrolytic efficiency of electrolyzer (b) in PV-BESS-PEM water electrolysis system.

hydrogen volume production only from 922.14 Nm^3 to 1002.63 Nm^3, which is an increase of only 8.73 %. In addition, when the water temperature is increased by a certain temperature, the effect on the hydrogen production of the system is very small, for example, the volume of hydrogen production hardly changes when the water temperature is changed from 350.15 K to 353.15 K.

Fig. 16 (b) demonstrates the effect of different PV panel temperatures on the hydrogen production of the PV-BESS-PEM system. Fig. 16 (b) shows that the temperature of the PV panels has a relatively weak effect on the system hydrogen production. This is due to the BESS and the EMS. The temperature of the PV panels has an effect on the PV power, but the BESS sends outward electrical energy to guarantee the stable operation of the electrolysis unit. The effect of PV temperature on the system's hydrogen production was greater in the second half of the month than in the first half, due to the overall discharged state of the battery and the gradual decrease in the battery's ability to output electrical energy to the outside world, resulting in a more pronounced change in the hydrogen production.

4.2.3. Effect of membrane area

Fig. 17 presents the effect of the area of membrane on the hydrogen production and electrolysis efficiency of the PV-BESS-PEM system when the number of electrolytic cells is 70. From Fig. 17, we know that the hydrogen production and electrolysis efficiency vary monotonically and incrementally with the increase of the membrane edge length *Z* at different membrane edge lengths *Y*. However, when the side length *Z* of the membrane is constant, the hydrogen production and electrolysis efficiency of the system first increase and then decrease with the increase of the side length *Y* of the membrane. The hydrogen production and electrolytic efficiency increased when the side length *Y* of the membrane was varied from 100 cm to 125 cm, and decreased when the side length *Y* of the membrane was varied from 125 cm to 130 cm. In addition, the overall effect of membrane area on system performance is less pronounced than the effects of number of electrolytic cells and temperature.

4.2.4. Effect of membrane thickness

Fig. 18 (a) and (b) show the effect of membrane thickness on hydrogen production and electrolysis efficiency of the PV-BESS-PEM water electrolysis system at an electrolysis cell number of 70, respectively. As shown in Fig. 18, the thickness of the membrane has almost no effect on the hydrogen production and electrolytic efficiency of the system. However, the increase of the thickness of the electrolyzer is helpful for the hydrogen production and electrolysis efficiency. When the thickness of the membrane was changed from 0.5 cm to 2 cm, the hydrogen production changed by about 0.25 kg, which is negligible, and the electrolysis efficiency changed by only about 0.16 %.

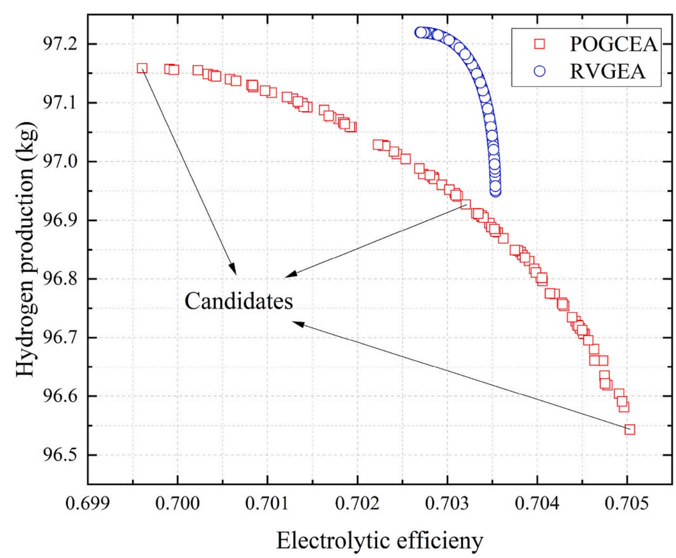

Fig. 19. Pareto front of optimization problem.

4.3. Optimization results

Two algorithms, POGCEA and RVGEA, are used to drive the optimization process, and the optimization results are shown in Fig. 19. From Fig. 19, it is not difficult to find that the optimization results using POGCEA are more uniformly distributed in space and perform better than RVGEA. There are three candidate solutions in the pareto frontier of POGCEA, considering the performance of the two optimization objectives, this study chooses the middle candidate point as the optimal solution.

Table 6 shows the comparison between the initial solution and the optimal solution. From Table 6, it can be seen that the hydrogen production of the optimized PV-BESS-PEM electrolytic water system is increased by 16.80 % and the electrolysis efficiency of the PEM electrolyzer is increased by 12.08 %. Fig. 20 shows the collected heating capacity and electricity consumption of an office building heating system in January 2022. After calculation, the optimal hydrogen production of the PV-BESS-PEM water electrolysis system obtained in this paper can fully satisfy the equivalent heat value of the office building's

Table 6
Comparison of the initial solution with the optimal solution.

	x_1	x_2	x_3	x_4	x_5	Hydrogen production (kg)	Electrolytic efficiency (%)
Initial	70	343.15	100.00	100.00	2.00	82.99	62.77
Optimal	80	347.75	113.90	125.00	1.36	96.93	70.35
Improvement	/	/	/	/	/	+16.80 %	+12.08 %

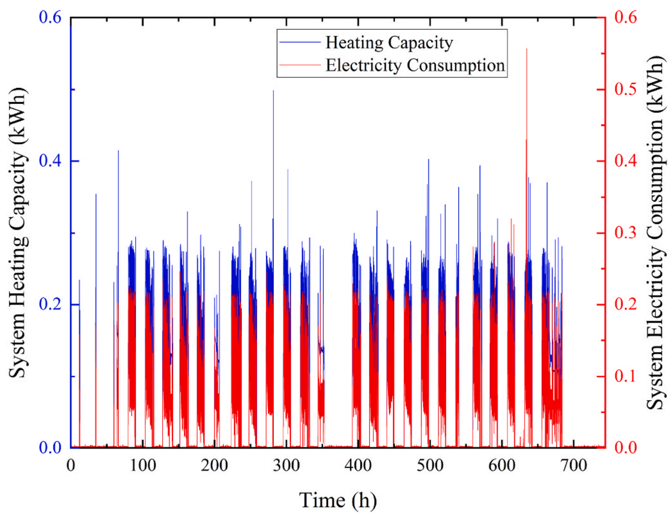

Fig. 20. The heating capacity and electricity consumed by the heating system of an office building.

electricity consumption.

5. Conclusions

In this study, an off-grid PV-BESS-PEM water electrolysis system is established based on Matlab/Simulink, and the accuracy of the model is verified on the basis of modeling the system. The main conclusions of this study are as follows:

(1) When the solar irradiance is strong, the excess PV power is stored by the BESS. Conversely, when the solar irradiance is weak, the BESS discharges electricity while satisfying hydrogen production. The proposed EMS can maintain the voltage stability of the DC bus and guarantee the continuous and stable operation of the hydrogen production process.
(2) The sensitivity analysis of the PV-BESS-PEM water electrolysis system reveals the influence of the number of electrolyzer cells, the water temperature, the membrane area and the membrane thickness on the hydrogen production and the electrolysis efficiency. Compared with the number of 50 cells in the electrolyzer, the hydrogen production of the increased by 64.00 % when the number of 80 cells in the electrolyzer. The change in water temperature from 341.15 K to 344.15 K increases the hydrogen volume production from 567.82 Nm^3 to 744.92 Nm^3. When the thickness of the membrane was changed from 0.5 cm to 2 cm, the electrolysis efficiency changed by only about 0.16 %.
(3) For surrogate models, the modeling accuracy of different machine learning methods varies across problems. The accuracy of the GPR model is higher for the first optimization objective and the accuracy of the WAS model is higher for the second optimization objective. In addition, the POGCEA shows superiority over RVGEA in solving the optimization model in this study.
(4) Compared to the initial solution, the optimal solution of the water electrolysis system has increased the hydrogen production by 16.80 % and the electrolysis efficiency of the PEM electrolyzer by 12.08 %, which indicates that the proposed method also contributes to the improvement of the energy utilization of the whole system.

There are still some limitations for this paper, and the following research can be continued in the future.

(1) In this paper, an off-grid renewable energy-based hydrogen production system is proposed, and machine learning and multi-objective strategies are combined to optimize the system. However, how to combine renewable energy sources (e.g., solar, wind) with conventional power systems, and then realize adaptive control based on artificial intelligence to improve energy efficiency and reliability is one of the future works to be carried out.
(2) The control method used in this paper can effectively simulate the operation of the system. However, it is possible to improve the control efficiency and increase the hydrogen production by employing more advanced controllers to optimize the charging and discharging times of BESS based machine learning. The evaluation of the potential benefits of hybrid renewable energy hydrogen production systems and the development of other control strategies are also worthy of exploration in the future.

CRediT authorship contribution statement

Ningbo Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yanhua Guo:** Software, Methodology, Data curation, Conceptualization. **Lu Liu:** Writing – review & editing, Supervision, Conceptualization. **Shuangquan Shao:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 52076085) and the program for HUST Academic Frontier Youth Team (No. 2019QYTD10). All developed codes and models supporting the findings of this study are available from the authors upon reasonable request.

References

[1] N. Wang, C. Li, W. Li, X. Chen, Y. Li, D. Qi, Heat dissipation optimization for a serpentine liquid cooling battery thermal management system: an application of surrogate assisted approach, J. Energy Storage 40 (2021) 102771.
[2] U.S. Energy Information Administration, International Energy Outlook 2022 (2021). https://www.eia.gov/outlooks/ieo/.
[3] J. Zhang, B. Ling, Y. He, Y. Zhu, Z. Wang, Life cycle assessment of three types of hydrogen production methods using solar energy, Int. J. Hydrogen Energy 47 (30) (2022) 14158–14168.

[4] O. Arslan, E. Acikkalp, G. Genc, A multi-generation system for hydrogen production through the high-temperature solid oxide electrolyzer integrated to 150 MW coal-fired steam boiler, Fuel 315 (2022) 123201.

[5] H. Lee, B. Lee, M. Byun, H. Lim, Economic and environmental analysis for PEM water electrolysis based on replacement moment and renewable electricity resources, Energy Convers. Manag. 224 (2020) 113477.

[6] M. İnci, Technoeconomic analysis of fuel cell vehicle-to-grid (FCV2G) system supported by photovoltaic energy, Energy Technol. 11 (4) (2023) 2201162.

[7] P. Nikolaidis, A. Poullikkas, A comparative overview of hydrogen production processes, Renew. Sustain. Energy Rev. 67 (2017) 597–611.

[8] F. Xue, J. Su, P. Li, Y. Zhang, Application of proton exchange membrane electrolysis of water hydrogen production technology in power plant, IOP Conf. Ser. Earth Environ. Sci. 631 (1) (2021) 012079.

[9] D.D.A. de Fátima Palhares, L.G.M. Vieira, J.J.R. Damasceno, Hydrogen production by a low-cost electrolyzer developed through the combination of alkaline water electrolysis and solar energy use, Int. J. Hydrogen Energy 43 (9) (2018) 4265–4275.

[10] H. Tebibel, Off grid PV system for hydrogen production using PEM methanol electrolysis and an optimal management strategy, Int. J. Hydrogen Energy 42 (30) (2017) 19432–19445.

[11] Y. Kikuchi, T. Ichikawa, M. Sugiyama, M. Koyama, Battery-assisted low-cost hydrogen production from solar energy: rational target setting for future technology systems, Int. J. Hydrogen Energy 44 (3) (2019) 1451–1465.

[12] M. Nasser, T.F. Megahed, S. Ookawara, H. Hassan, A review of water electrolysis–based systems for hydrogen production using hybrid/solar/wind energy systems, Environ. Sci. Pollut. Control Ser. 29 (58) (2022) 86994–87018.

[13] N. Li, S.S. Araya, S.K. Kær, Investigating low and high load cycling tests as accelerated stress tests for proton exchange membrane water electrolysis, Electrochim. Acta 370 (2021) 137748.

[14] F. Sayedin, A. Maroufmashat, S. Sattari, A. Elkamel, M. Fowler, Optimization of Photovoltaic Electrolyzer Hybrid systems; taking into account the effect of climate conditions, Energy Convers. Manag. 118 (2016) 438–449.

[15] Z. Yang, J. Lin, H. Zhang, B. Lin, G. Lin, A new direct coupling method for photovoltaic module-PEM electrolyzer stack for hydrogen production, Fuel Cell. 18 (4) (2018) 543–550.

[16] A. Mraoui, B. Benyoucef, L. Hassaine, Experiment and simulation of electrolytic hydrogen production: case study of photovoltaic-electrolyzer direct connection, Int. J. Hydrogen Energy 43 (6) (2018) 3441–3450.

[17] X. Cai, R. Lin, J. Xu, Y. Lu, Construction and analysis of photovoltaic directly coupled conditions in PEM electrolyzer, Int. J. Hydrogen Energy 47 (10) (2022) 6494–6507.

[18] A. Maroufmashat, F. Sayedin, S.S. Khavas, An imperialist competitive algorithm approach for multi-objective optimization of direct coupling photovoltaic-electrolyzer systems, Int. J. Hydrogen Energy 39 (33) (2014) 18743–18757.

[19] B. Laoun, A. Khellaf, M.W. Naceur, A.M. Kannan, Modeling of solar photovoltaic-polymer electrolyte membrane electrolyzer direct coupling for hydrogen generation, Int. J. Hydrogen Energy 41 (24) (2016) 10120–10135.

[20] S. Wang, X. Wu, S. Jafarmadar, P.K. Singh, S. Khorasani, M. Marefati, et al., Numerical assessment of a hybrid energy system based on solid oxide electrolyzer, solar energy and molten carbonate fuel cell for the generation of electrical energy and hydrogen fuel with electricity storage option, J. Energy Storage 54 (2022) 105274.

[21] A. Ibáñez-Rioja, P. Puranen, L. Järvinen, A. Kosonen, V. Ruuskanen, J. Ahola, et al., Simulation methodology for an off-grid solar–battery–water electrolyzer plant: simultaneous optimization of component capacities and system control, Appl. Energy 307 (2022) 118157.

[22] P. Puranen, A. Kosonen, J. Ahola, Technical feasibility evaluation of a solar PV based off-grid domestic energy system with battery and hydrogen energy storage in northern climates, Sol. Energy 213 (2021) 246–259.

[23] X. Cao, J. Wang, P. Zhao, H. Xia, Y. Li, L. Sun, et al., Hydrogen production system using alkaline water electrolysis adapting to fast fluctuating photovoltaic power, Energies 16 (8) (2023) 3308.

[24] A. Cecilia, J. Carroquino, V. Roda, R. Costa-Castello, F. Barreras, Optimal energy management in a standalone microgrid, with photovoltaic generation, short-term storage, and hydrogen production, Energies 13 (6) (2020) 1454.

[25] F. Gutiérrez-Martín, L. Amodio, M. Pagano, Hydrogen production by water electrolysis and off-grid solar PV, Int. J. Hydrogen Energy 46 (57) (2021) 29038–29048.

[26] A. Izadi, M. Shahafve, P. Ahmadi, Neural network genetic algorithm optimization of a transient hybrid renewable energy system with solar/wind and hydrogen storage system for zero energy buildings at various climate conditions, Energy Convers. Manag. 260 (2022) 115593.

[27] G. Li, J. Li, Z. Dai, Akram M. Waqar, Modelling and analysis of a novel hydrogen production approach by full spectrum solar energy, Energy Convers. Manag. 263 (2022) 115694.

[28] G. Li, J. Li, R. Yang, X. Chen, Performance analysis of a hybrid hydrogen production system in the integrations of PV/T power generation electrolytic water and photothermal cooperative reaction, Appl. Energy 323 (2022) 119625.

[29] M. Marefati, M. Mehrpooya, Introducing a hybrid photovoltaic solar, proton exchange membrane fuel cell and thermoelectric device system, Sustain. Energy Technol. Assessments 36 (2019) 100550.

[30] A. Allouhi, Management of photovoltaic excess electricity generation via the power to hydrogen concept: a year-round dynamic assessment using Artificial Neural Networks, Int. J. Hydrogen Energy 45 (41) (2020) 21024–21039.

[31] A.K. Pathak, R. Kothari, V.V. Tyagi, S. Anand, Integrated approach for textile industry wastewater for efficient hydrogen production and treatment through solar PV electrolysis, Int. J. Hydrogen Energy 45 (48) (2020) 25768–25782.

[32] M. İnci, K.Ç. Bayındır, Single-stage vehicular fuel cell system with harmonic elimination capability to suppress distortion effects of electric vehicle parking lots, J. Power Sources 597 (2024) 234175.

[33] M. İnci, Connecting multiple vehicular PEM fuel cells to electrical power grid as alternative energy sources: a Case Study, Int. J. Hydrogen Energy 52 (2024) 1035–1051.

[34] G. Perkins, Techno-economic comparison of the levelised cost of electricity generation from solar PV and battery storage with solar PV and combustion of bio-crude using fast pyrolysis of biomass, Energy Convers. Manag. 171 (2018) 1573–1588.

[35] X. Gu, Z. Ying, X. Zheng, B. Dou, G. Cui, Photovoltaic-based energy system coupled with energy storage for all-day stable PEM electrolytic hydrogen production, Renew. Energy 209 (2023) 53–62.

[36] A. Poullikkas, A comparative overview of large-scale battery systems for electricity storage, Renewable Sustainable Energy Rev. 27 (2013) 778–788.

[37] M. Koller, T. Borsche, A. Ulbig, G. Andersson, Review of grid applications with the Zurich 1MW battery energy storage system, Elec. Power Syst. Res. 120 (2015) 128–135.

[38] A. Ibáñez-Rioja, L. Järvinen, P. Puranen, A. Kosonen, V. Ruuskanen, K. Hynynen, et al., Off-grid solar PV–wind power–battery–water electrolyzer plant: simultaneous optimization of component capacities and system control, Appl. Energy 345 (2023) 121277.

[39] F. Barbir, PEM electrolysis for production of hydrogen from renewable energy sources, Sol. Energy 78 (5) (2005) 661–669.

[40] R. Garcia-Valverde, N. Espinosa, A. Urbina, Simple PEM water electrolyser model and experimental validation, Int. J. Hydrogen Energy 37 (2) (2012) 1927–1938.

[41] F.Z. Aouali, M. Becherif, H.S. Ramadan, M. Emziane, A. Khellaf, K. Mohammedi, Analytical modelling and experimental validation of proton exchange membrane electrolyser for hydrogen production, Int. J. Hydrogen Energy 42 (2) (2017) 1366–1374.

[42] D.S. Falcão, A.M.F.R. Pinto, A review on PEM electrolyzer modelling: guidelines for beginners, J. Clean. Prod. 261 (2020) 121184.

[43] S. Wang, X. Wu, S. Jafarmadar, P.K. Singh, S. Khorasani, M. Marefati, et al., Numerical assessment of a hybrid energy system based on solid oxide electrolyzer, solar energy and molten carbonate fuel cell for the generation of electrical energy and hydrogen fuel with electricity storage option, J. Energy Storage 54 (2022) 105274.

[44] J. Zhao, X. Luo, Z. Tu, S.H. Chan, A novel CCHP system based on a closed PEMEC-PEMFC loop with water self-supply, Appl. Energy 338 (2023) 120921.

[45] N. Wang, C. Li, W. Li, M. Huang, D. Qi, Effect analysis on performance enhancement of a novel air cooling battery thermal management system with spoilers, Appl. Therm. Eng. 192 (2021) 116932.

[46] C. He, R. Cheng, Y. Tian, X. Zhang, K.C. Tan, Y. Jin, Paired offspring generation for constrained large-scale multiobjective optimization, IEEE Trans. Evol. Comput. 25 (3) (2021) 448–462.

[47] R. Cheng, Y. Jin, M. Olhofer, B. Sendhoff, A reference vector guided evolutionary algorithm for many-objective optimization, IEEE Trans. Evol. Comput. 20 (5) (2016) 773–791.

[48] J. Page, Chapter II-1-A - the role of solar-radiation climatology in the design of photovoltaic systems, in: S.A. Kalogirou (Ed.), McEvoy's Handbook of Photovoltaics, third ed., Academic Press, 2018, pp. 601–670.